

V6.2 · Shared Biochemical Motifs

Cross-theme overlap is represented, not suppressed

Objective	Identify motifs whose chemistry genuinely spans more than one theme, and explain each overlap spectroscopically.
Method	A motif is flagged as shared when its dominant L2 membership is below 0.85. For each, the responsible components and the overlapping band positions are reported.
Frozen	Atlas 09ed804a40836f4a05a91ba10900cded; the V6 MSS map M is an input, never rewritten.
Reproduce	results/v6_rebuild/code/p10 → p11 → p12 → p13

V6 forced *protein_amide_backbone* and the saccharide motifs apart into disjoint themes. That was a modelling decision fighting the spectroscopy: amide III (~1240) and the CH₂/C-O saccharide modes (~1122-1340) genuinely overlap. V6.2 represents the overlap.

Part 3 — shared motifs are a feature, not a defect

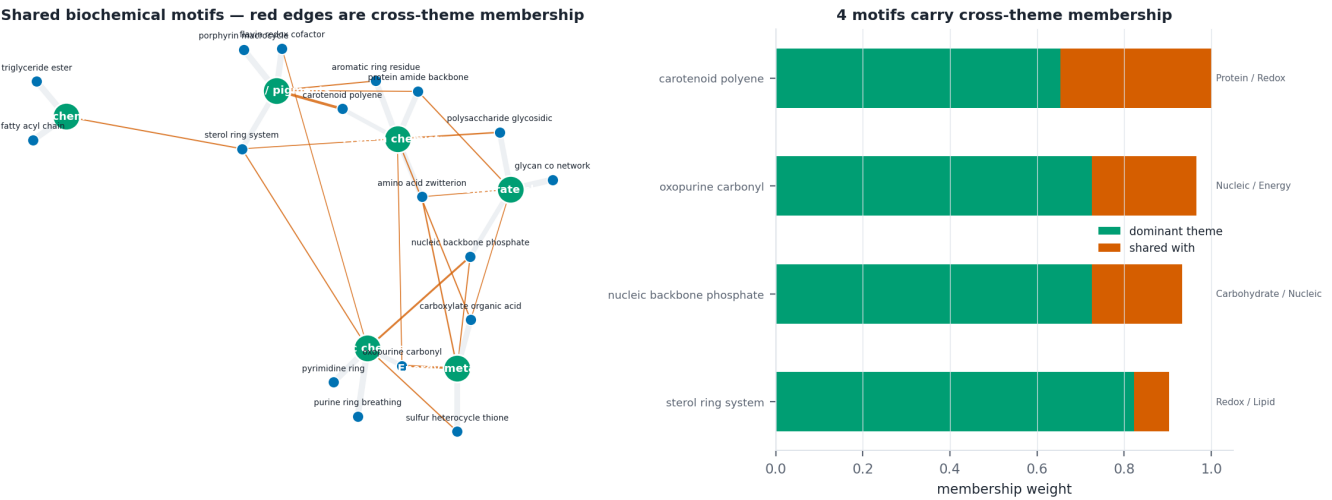


Figure 1 — The motif→theme network (red edges are cross-theme membership) and the shared-motif inventory.

Results

motif	dominant	w	shared with	w	components	shared bands cm ⁻¹
carotenoid polyene	Protein chemistry	0.654	Redox / pigments	0.346	c6(0.41), c12(0.24), c5(0.20)	1005, 1157, 1190, 1270, 1520
oxopurine carbonyl	Nucleic chemistry	0.725	Energy metabolism	0.240	c15(0.33), c19(0.21), c20(0.14)	640, 938
nucleic backbone phosphate	Carbohydrate chemistry	0.726	Nucleic chemistry	0.207	c17(0.21), c23(0.18), c4(0.15)	788, 920, 1240
sterol ring system	Redox / pigments	0.823	Lipid chemistry	0.079	c3(0.22), c8(0.21), c13(0.17)	548, 608, 702, 958, 1128, 1440

Interpretation

4 of 17 motifs carry meaningful cross-theme membership. Each has a spectroscopic reason on the same row: *nucleic_backbone_phosphate* shares the 788/920/1240 cm^{-1} region with the nucleobases; *oxopurine_carbonyl* shares 640/938 with the organic acids through its carbonyl; *carotenoid_polyene* shares the 1005–1520 region with aromatic residues because both are conjugated π systems. These are not confusions to be fixed — they are the reason a Raman spectrum of a mixture is hard, stated explicitly.

This matters for transfer. A hierarchy that hides overlap will mis-attribute it on a new substrate; one that represents it can carry the ambiguity forward into the Ag-SERS phase as an explicit prior.

Limitations

- The 0.85 threshold for 'shared' is a reporting convention, not a test.
- *sterol_ring_system* appears as shared with Lipid chemistry only weakly (0.08) because its dominant assignment is already wrong — the shared-motif view does not rescue it.
- Overlap is measured on this corpus; a corpus with more phospholipids would likely surface further sharing.

Recommendations

- Propagate the shared membership into the BSV rather than collapsing to the dominant theme, so downstream consumers see the ambiguity.
- Use the shared-motif list as the prior for which themes will be hardest to separate on silver.

Atlas 09ed804a40836f4a05a91ba10900cded verified unchanged. All values recomputed from results/v6_rebuild/.